

# Consensus without Accuracy: Investigating LLMs Recovery of Item Difficulty Using Paired Comparisons

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## Abstract

This study evaluates whether large language models (LLMs) can recover item difficulty estimates through Bradley-Terry modeling from pairwise comparisons of items. Across five Item Response Warehouse datasets and four LLMs, we examine alignment between LLM pairwise-derived difficulty rankings and 1PL IRT parameters, with implications for scalable, AI-assisted item calibration.

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## Structured Summary

### Background

Item calibration is central to educational assessment but resource-intensive, typically requiring large samples of student response data and extended field-testing periods. Recent advances in large language model (LLM) capabilities raise an intriguing question: can item difficulty be meaningfully estimated directly from item text, without needing response data? Such an approach would offer immediate utility for item writers and test developers seeking a scalable first-pass difficulty signal for item screening, ordering, or rough-cut form assembly before formal calibration. Prior work has explored LLMs for automated item generation and scoring (von Davier et al., 2022), but their deployment as a calibration proxy for estimating the psychometric difficulty of items never yet administered remains less examined as does the extent to which paired comparisons aid in that effort.

### Aims

This study addresses three questions:

- 1) To what extent do Bradley-Terry scores derived from LLM pairwise comparisons correlate with 1PL difficulty parameters across educational item sets?
- 2) How consistent and reliable are LLM pairwise judgments?
- 3) Do results vary systematically across LLMs and item domains?

### Sample(s)

Items were drawn from five datasets in the Item Response Warehouse (Domingue et al., 2025): frac20 - a mathematics/fractions dataset (Tatsuoka, 2002) - and four literacy-focused datasets: gilbert\_meta\_102, gilbert\_meta\_103, and gilbert\_meta\_104 (Relyea et al., 2025) as well as gilbert\_meta\_2 (Kim et al., 2023). These datasets include items with established 1PL IRT difficulty calibrations, providing a psychometric ground truth for validation. Four LLMs served as raters: GPT-5.4, GPT-5.4 Mini, Gemini 3 Flash, and Gemini 3.1 Pro.

## Methods

For each item, a four-prompt Concept-Guided Chain-of-Thought (CGCoT) chain was administered sequentially to the LLM to target cognitive item difficulty (Wu et al., 2024). After the LLM answers each prompt, its response is concatenated to all its previous responses and fed into the next prompt until at the end an entire breakdown of the item is created, as displayed in **Figure 1**. The resulting item breakdowns are used to generate comparisons across item pairs within each dataset. Based on the cumulative output from these four steps applied to an item, the LLM judged which item in a pair required greater latent cognitive ability for correct response.

Bradley-Terry (BT) (Bradley & Terry, 1952) models were fitted to each LLM's choices for each dataset to produce a difficulty score for each item using pairadigm (Chrzan, 2026). A Hierarchical Bradley-Terry (HBT) model (Böckenholt, 2001) was also fit to assess each LLM's reliability and position bias. Additionally, triplet transitivity assessed model internal consistency. Inter-rater agreement across LLMs was estimated using Krippendorff's Alpha to assess the consistency across models. Finally, convergent validity was assessed via Spearman correlations between each model's BT scores and the corresponding 1PL difficulty parameters.

## Preliminary Results

Overall Spearman correlations between BT scores and 1PL difficulty parameters varied across models (**Table 1**). Gemini 3.1 Pro produced the strongest alignment, followed by the HBT ensemble and Gemini 3 Flash. GPT-5.4 yielded the poorest alignment overall (**Figure 2**). Dataset-stratified results (**Table 2, Figure 3**) reveal meaningful heterogeneity across datasets. In particular, performance on gilbert\_meta\_104 - the smallest dataset - was weaker across individual models, though the HBT model still performed well suggesting that ensembling partially compensates for small-N instability. GPT-5.4 produced near-zero or negative correlations on two of the five datasets, a pattern inconsistent with random noise and suggestive of systematic misranking.

Despite the modest convergent validity observed, both internal consistency metrics indicated high agreement. Alpha values across datasets ranged from 0.848 to 0.901, indicating consensus in difficulty rankings across all four models (**Table 3**). Transitivity was similarly high, with most model-dataset combinations exceeding 98% (**Table 4**). These levels of transitivity mirror those found in empirical studies of human comparison behavior which show high transitivity (Sleboda & Sokolowska, 2017). Critically, this high agreement coexisted with the weak and negative convergent validity observed for GPT-5.4. The models were reliably reaching consensus while simultaneously being wrong with respect to psychometric ground truth.

**Figure 4** presents HBT-estimated reliability (alpha) and position bias (gamma) by model and dataset. Reliability estimates were generally modest, with wide 94% HDI intervals reflecting instability attributable to small numbers of items in each dataset. GPT-5.4 Mini showed the highest alpha estimate on the frac20 dataset, while Gemini models tended toward slightly higher reliability on several literacy datasets. Position bias was close to zero for most model-dataset combinations in frac20, but showed larger and more variable estimates in the Gilbert datasets.

## Conclusions / Anticipated Contributions

These findings have meaningful implications for the use of LLMs in AI-assisted item calibration and assessment design. On the positive side, LLM pairwise comparisons do recover a meaningful, if moderate, signal about item difficulty ordering and stronger frontier models consistently outperforming smaller ones. This suggests potential utility as a low-cost screening tool for rough difficulty ordering early in item development.

Importantly, however, model size and cost do not straightforwardly predict validity. Our results suggest that sensitivity to the judgment task may vary meaningfully across model families in ways that are not reducible to scale, and that practitioners still must empirically evaluate models rather than defaulting to the largest or most recent option available.

The results also carry an important caution: high inter-rater reliability and internal consistency among LLMs are insufficient evidence of performance. The near-perfect triplet transitivity and high alphas observed here could easily be mistaken for evidence that LLM judgments are trustworthy, when in fact the models may be consistently wrong in correlated ways. Consensus is not accuracy.

Finally, the HBT model consistently ranked among the strongest performers, achieving the strongest correlations on several datasets. This suggests that aggregating across LLM raters may be a practical strategy for improving performance without requiring access to a stronger individual model, and warrants further investigation. To this end, future work could examine the ability to use an ensemble of local LLMs or encoder models, which have shown more promise in their ability to detect item difficulty than seen here (Kapoor et al., 2025). Further, as the IRW Item Text data expands, we intend to expand our analysis to more datasets.

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**Table 1: Convergent Validity Metrics for LLM-Derived Difficulty Scores by Model**

| Model          | Spearman $\rho$ | p-value | N<br>(number of items) |
|----------------|-----------------|---------|------------------------|
| GPT-5.4        | 0.162           | 0.084   | 114                    |
| GPT-5.4 Mini   | 0.231           | 0.013   | 114                    |
| Gemini 3 Flash | 0.414           | < 0.001 | 114                    |
| Gemini 3.1 Pro | <b>0.484</b>    | < 0.001 | 114                    |
| HBT            | 0.445           | < 0.001 | 114                    |

*Note: Best performing model bolded*

**Table 2: Convergent Validity Metrics for LLM-Derived Difficulty Scores by Model and Dataset**

| Dataset                 | Model                 | Spearman $\rho$ | p-value           | N         |
|-------------------------|-----------------------|-----------------|-------------------|-----------|
| <i>frac20</i>           | GPT-5.4               | 0.355           | 0.125             | 20        |
|                         | GPT-5.4 Mini          | 0.411           | 0.072             | 20        |
|                         | Gemini 3 Flash        | 0.293           | 0.210             | 20        |
|                         | <b>Gemini 3.1 Pro</b> | <b>0.537</b>    | <b>0.015</b>      | <b>20</b> |
|                         | HBT                   | 0.474           | <0.001            | 20        |
| <i>gilbert_meta_102</i> | GPT-5.4               | -0.092          | 0.595             | 36        |
|                         | GPT-5.4 Mini          | 0.233           | 0.172             | 36        |
|                         | <b>Gemini 3 Flash</b> | <b>0.604</b>    | <b>&lt; 0.001</b> | <b>36</b> |
|                         | Gemini 3.1 Pro        | 0.493           | 0.002             | 36        |
|                         | HBT                   | 0.537           | <0.001            | 36        |
| <i>gilbert_meta_103</i> | GPT-5.4               | 0.327           | 0.084             | 29        |
|                         | GPT-5.4 Mini          | 0.140           | 0.469             | 29        |
|                         | Gemini 3 Flash        | 0.349           | 0.063             | 29        |
|                         | <b>Gemini 3.1 Pro</b> | <b>0.501</b>    | <b>0.006</b>      | <b>29</b> |
|                         | HBT                   | 0.423           | <0.001            | 29        |
| <i>gilbert_meta_104</i> | GPT-5.4               | -0.033          | 0.932             | 9         |
|                         | GPT-5.4 Mini          | -0.083          | 0.831             | 9         |

| Dataset               | Model                 | Spearman $\rho$ | p-value          | N         |
|-----------------------|-----------------------|-----------------|------------------|-----------|
|                       | Gemini 3 Flash        | 0.500           | 0.170            | 9         |
|                       | Gemini 3.1 Pro        | 0.533           | 0.139            | 9         |
|                       | <b>HBT</b>            | <b>0.583</b>    | <b>&lt;0.001</b> | <b>9</b>  |
| <i>gilbert_meta_2</i> | GPT-5.4               | 0.368           | 0.110            | 20        |
|                       | GPT-5.4 Mini          | 0.355           | 0.125            | 20        |
|                       | Gemini 3 Flash        | 0.298           | 0.202            | 20        |
|                       | <b>Gemini 3.1 Pro</b> | <b>0.409</b>    | <b>0.073</b>     | <b>20</b> |
|                       | HBT                   | 0.392           | <0.001           | 20        |

*Note: Highest correlation model within each dataset is bolded.*

**Table 3: Krippendorff’s Alpha Across LLMs by Dataset**

| Dataset                 | Krippendorff’s Alpha |
|-------------------------|----------------------|
| <i>gilbert_meta_2</i>   | 0.893                |
| <i>frac20</i>           | 0.858                |
| <i>gilbert_meta_102</i> | 0.849                |
| <i>gilbert_meta_103</i> | 0.901                |
| <i>gilbert_meta_104</i> | 0.848                |

**Table 4: Triplet Transitivity by Model and Dataset**

| Dataset               | Model                 | Transitivity (%) | Triplets (N) |
|-----------------------|-----------------------|------------------|--------------|
| <i>gilbert_meta_2</i> | <b>Gemini 3 Flash</b> | <b>100.00%</b>   | <b>644.0</b> |
|                       | <b>Gemini 3.1 Pro</b> | <b>100.00%</b>   | <b>644.0</b> |
|                       | GPT-5.4 Mini          | 95.96%           | 644.0        |
|                       | GPT-5.4               | 98.91%           | 644.0        |
| <i>frac20</i>         | <b>Gemini 3 Flash</b> | <b>99.69%</b>    | <b>637.0</b> |
|                       | Gemini 3.1 Pro        | 99.06%           | 637.0        |
|                       | GPT-5.4 Mini          | 99.37%           | 637.0        |
|                       | GPT-5.4               | 97.33%           | 637.0        |

| Dataset                 | Model                 | Transitivity (%) | Triplets (N) |
|-------------------------|-----------------------|------------------|--------------|
| <i>gilbert_meta_102</i> | <b>Gemini 3 Flash</b> | <b>99.57%</b>    | <b>719.0</b> |
|                         | Gemini 3.1 Pro        | 99.44%           | 719.0        |
|                         | GPT-5.4 Mini          | 93.88%           | 719.0        |
|                         | GPT-5.4               | 97.64%           | 719.0        |
| <i>gilbert_meta_103</i> | Gemini 3 Flash        | 98.99%           | 691.0        |
|                         | <b>Gemini 3.1 Pro</b> | <b>100.00%</b>   | <b>691.0</b> |
|                         | GPT-5.4 Mini          | 97.25%           | 691.0        |
|                         | GPT-5.4               | 99.42%           | 691.0        |
| <i>gilbert_meta_104</i> | <b>Gemini 3 Flash</b> | <b>100.00%</b>   | <b>84.0</b>  |
|                         | <b>Gemini 3.1 Pro</b> | <b>100.00%</b>   | <b>84.0</b>  |
|                         | GPT-5.4 Mini          | 98.81%           | 84.0         |
|                         | GPT-5.4               | 96.43%           | 84.0         |

*Note: Most transitive model within each dataset is bolded.*

Figure 1: Concept-Guided Chain-of-Thought Workflow

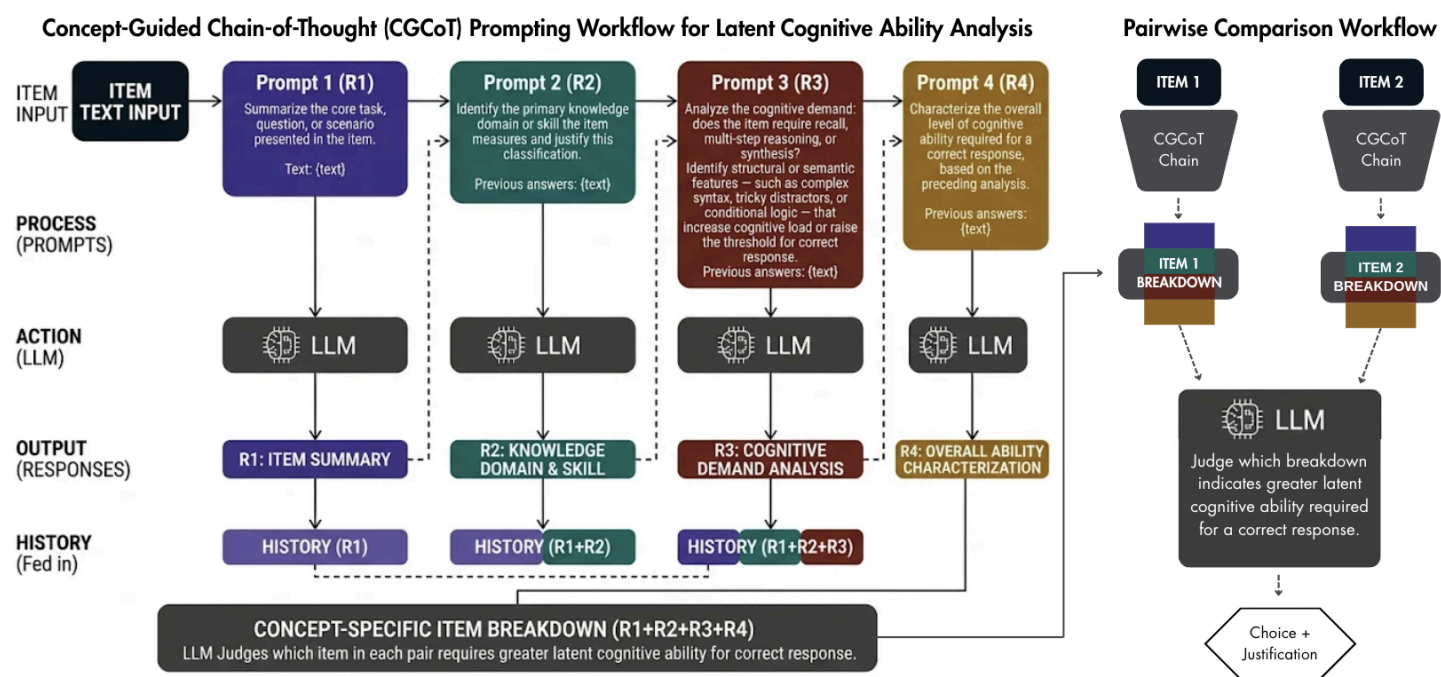


Figure 2: Bradley-Terry Scores versus 1PL Difficulty By Model

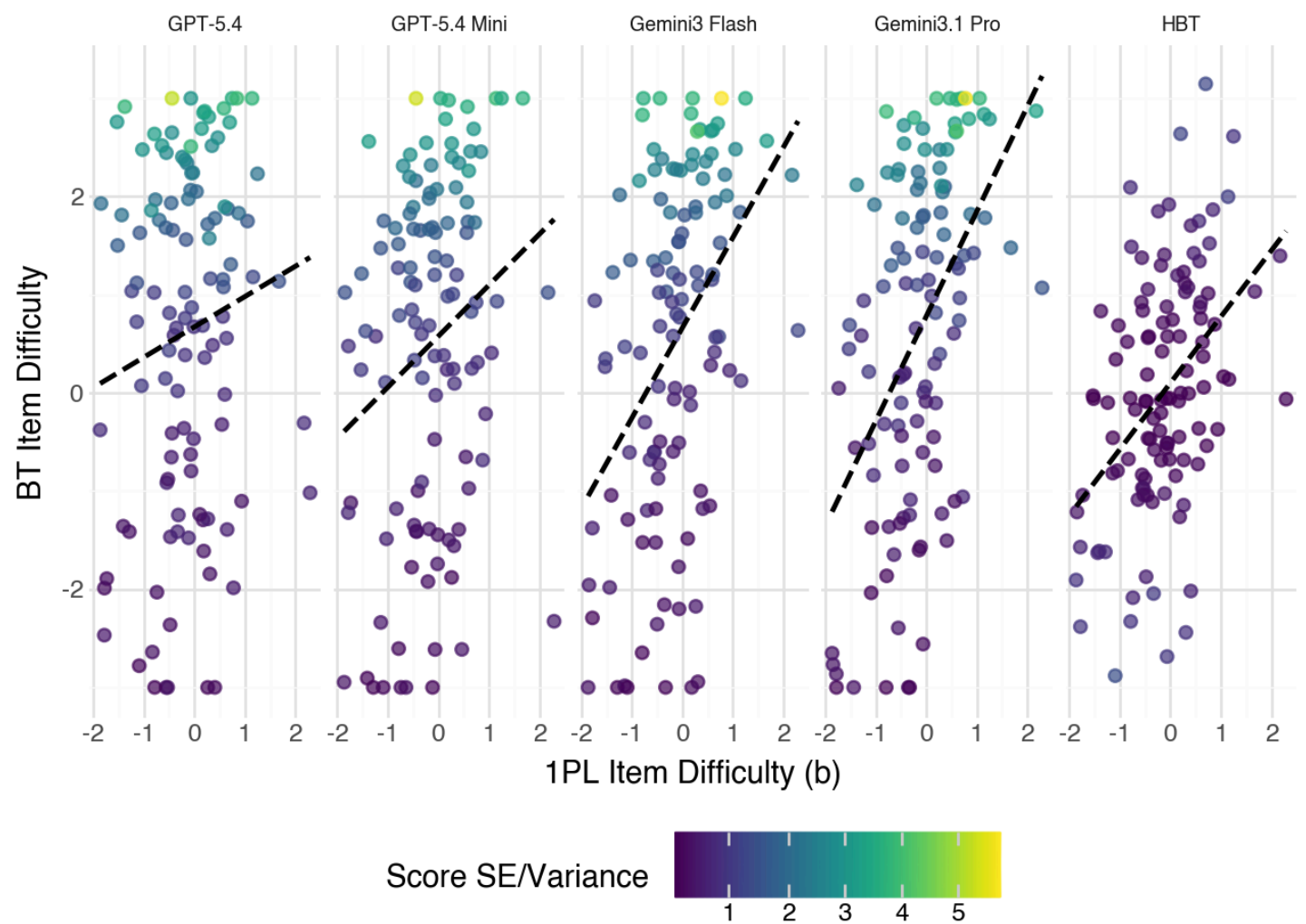


Figure 3: Bradley-Terry Scores versus 1PL Difficulty By Model and Dataset

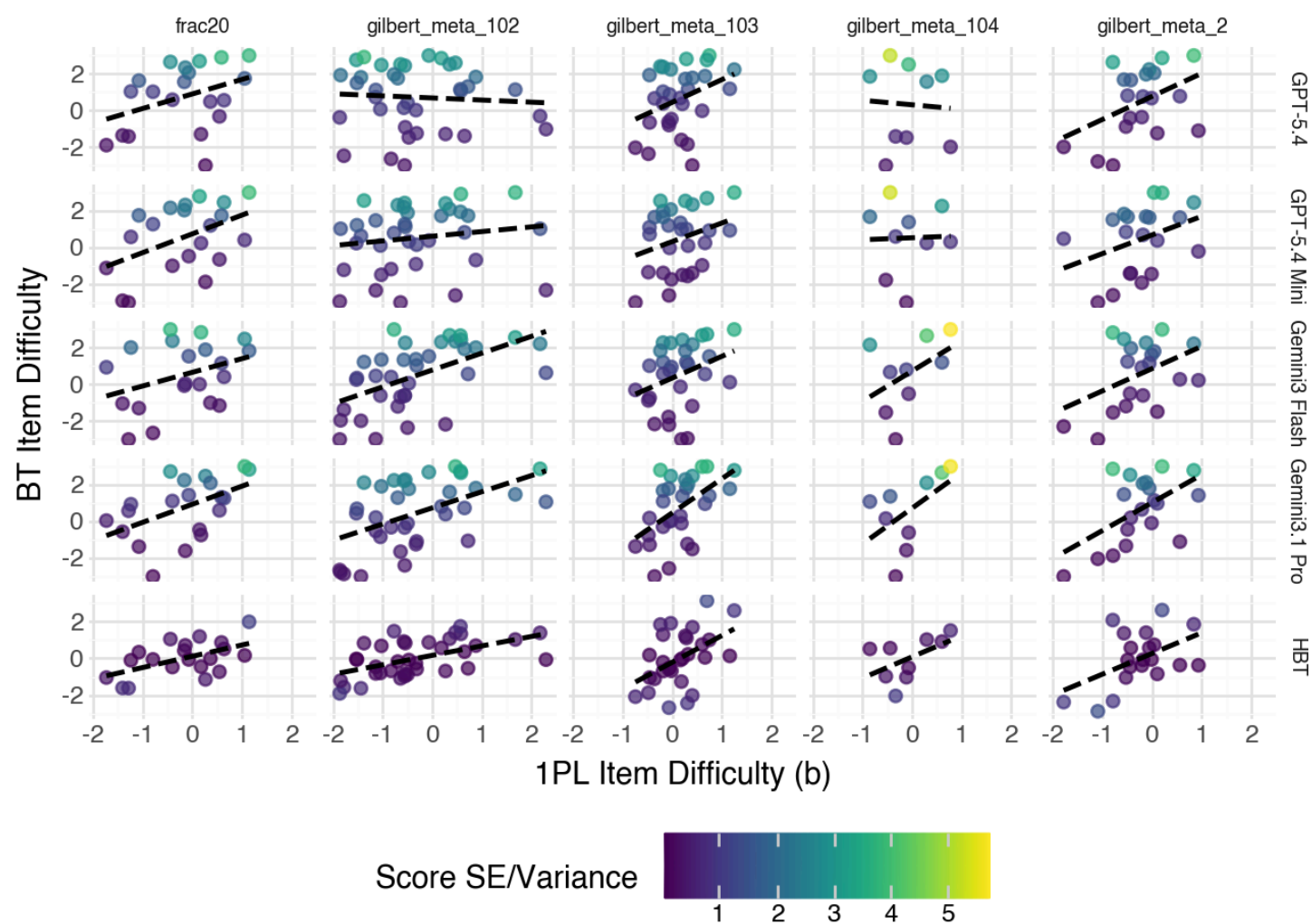


Figure 4: HBT Estimated LLM Reliability and Position Bias by Dataset

Errorbars indicate 94% HDI  
A. Higher alpha indicates stronger rater reliability  
B. Positive gamma indicates bias toward first item shown

